

पहली मंजिल — Architectural Design

First Floor Layout (Option F)

Building: Farm Storage / Farmhouse

Location: Mithanpur Urf Naimtulla Nagar, Moradabad, U.P.

Plot: Agricultural Farmland

Building Footprint: 20 ft (W) × 25 ft (D)

First Floor Height: 12 ft (floor to roof slab)

Date: June 2026

यह दस्तावेज़ पहली मंजिल का पूर्ण architectural design है।

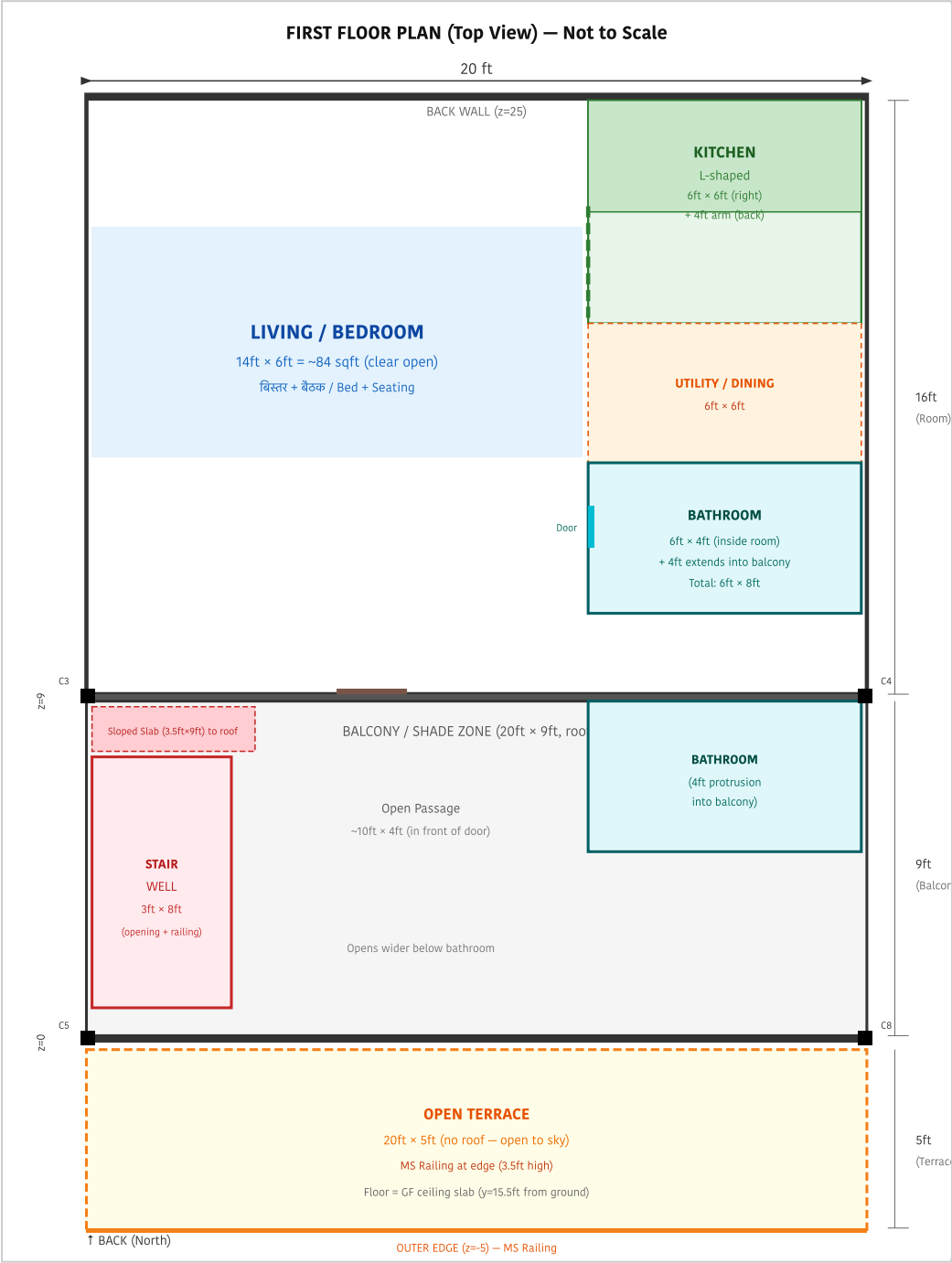
This document covers the complete first floor architectural design.

Print-ready • A4 format • For contractor reference

1. Floor Plan — पहली मंजिल का नक्शा

Total depth from back wall to terrace edge: 16ft (room) + 9ft (balcony/shade) + 5ft (open terrace)
= **30ft**

Width: **20ft** throughout

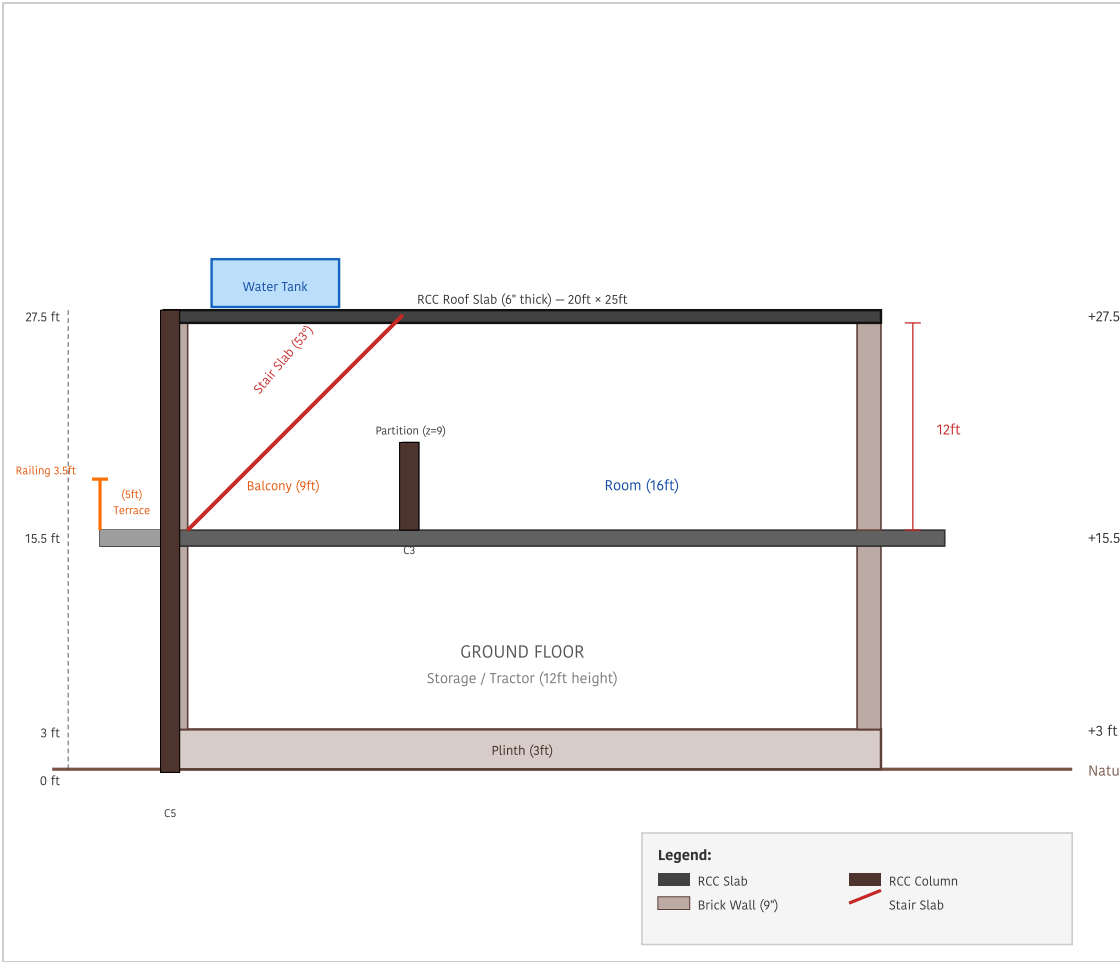


Note / नोट:

Room door opens from living area into balcony zone. Bathroom is en-suite (door from room only). Stair well provides access from ground floor. Sloped slab gives roof access at 53° angle.

2. Section View — Cross Section (Side Cut)

Looking from LEFT side (East-West cut through building center)



Height Summary / ऊँचाई सारांश

Level	Height from Ground	Description
Natural Ground	0 ft	जमीन का स्तर
Plinth / Platform	+3 ft	GF finished floor level
GF Ceiling / 1F Floor	+15.5 ft	RCC slab (6" thick) — GF 12ft clear + slab
1F Ceiling / Roof	+27.5 ft	RCC roof slab (6" thick) — 1F 12ft clear
Parapet Top	+30.5 ft	3ft parapet above roof slab
Water Tank Top	~+32 ft	Above stair zone on roof

3. Room-by-Room Description / कमरा विवरण

Space / स्थान	Dimensions	Area	Purpose / उपयोग	Floor Finish	Wall Finish
Living / Bedroom बैठक / शयनकक्ष	14ft × 6ft (clear open space)	~84 sqft	Main living area, bed placement, seating. No obstructions in this zone.	Vitrified tiles (2×2 ft, light color)	POP punning + emulsion paint
Kitchen रसोई (L-shaped)	6ft along right wall × 6ft deep + 4ft arm along back wall	~52 sqft	Cooking, storage, water supply. L-shape counter with gas, sink. Semi-open with 3.5ft half-wall.	Anti-skid ceramic tiles	Glazed tiles up to 4ft (dado), POP above
Bathroom शौचालय + स्नानगृह	6ft × 8ft total (4ft in room + 4ft in balcony)	48 sqft	WC (Indian style), shower, wash basin. En-suite access from room only.	Anti-skid ceramic tiles (dark)	Full height glazed tiles (7ft). Waterproofing on all walls.
Utility / Dining उपयोगिता / भोजन	6ft × 6ft	36 sqft	Small dining, fridge placement, washing machine, storage shelf.	Vitrified tiles	POP + emulsion paint
Balcony / Shade Zone बालकनी (छत वाली)	20ft × 9ft (minus stair & bath protrusion)	~130 sqft (net usable)	Covered passage, sitting area, clothes drying. Arrival point from GF stairs.	Rough textured tiles (non-slip)	External grade weather coat paint
Open Terrace खुली छत	20ft × 5ft	100 sqft	Open-to-sky sitting, evening relaxation, plants. MS railing at edge.	Rough kota stone / textured tiles	N/A (open sides, railing only)
Stair Well सीढ़ी	3ft × 8ft (opening in slab)	24 sqft	Access from GF. MS railing around opening. Head clearance: 7ft minimum.	N/A (opening)	Railing: MS pipe (1.5" dia)
Stair to Roof छत पर जाने की सीढ़ी	3.5ft × 9ft (sloped slab, 53°)	~31.5 sqft (slab area)	Access to roof for water tank maintenance and open terrace use.	MS ladder rungs / concrete steps on slab	N/A (open)

Total First Floor Area:

Room: $20\text{ft} \times 16\text{ft} = 320\text{ sqft}$ (carpet area)

Balcony (covered): $20\text{ft} \times 9\text{ft} = 180\text{ sqft}$

Terrace (open): $20\text{ft} \times 5\text{ft} = 100\text{ sqft}$

Total: 600 sqft (covered: 500 sqft)

4. Structural Notes / संरचना विवरण

4.1 Column Schedule / स्तंभ (Columns)

Column ID	Position	Size	Height (from plinth)	Remarks
C3	Left wall at z=9 (partition line)	9" × 12"	GF + 1F (plinth to roof = 24.5ft)	Supports partition beam + roof
C4	Right wall at z=9 (partition line)	9" × 12"	GF + 1F (plinth to roof = 24.5ft)	Supports partition beam + roof
C5	Front-left corner (z=0)	9" × 12"	GF + 1F (plinth to roof = 24.5ft)	Extended to roof for slab support. Visible on balcony.
C8	Front-right corner (z=0)	9" × 12"	GF + 1F (plinth to roof = 24.5ft)	Extended to roof for slab support. Visible on balcony.
C1, C2	Back wall corners (z=25)	9" × 12"	GF + 1F (plinth to roof = 24.5ft)	Back corners, support back beam

4.2 Beam Schedule / बीम (Beams)

Beam	Position	Size	Span	Purpose
Middle Deep Beam	z=9 (partition wall line)	9" × 20" (deep)	20 ft	Supports roof slab. Partition wall sits on this. Major structural element.
Back Deep Beam	z=25 (back wall)	9" × 20" (deep)	20 ft	Supports roof slab at back. Integrated with back wall.
Front Beam	z=0 (front wall)	9" × 15"	20 ft	Supports roof slab front edge. C5-C8 connection.
Side Beams (2)	Left & right walls	9" × 15"	25 ft	Along side walls, connect all columns

4.3 Slab Details

Slab	Thickness	Span	Reinforcement	Notes
Roof Slab (1F ceiling)	6" (150mm)	20ft × 25ft (building footprint)	Two-way slab, 10mm & 12mm bars	Covers entire building. Does NOT extend over terrace. Waterproofing on top.
GF Ceiling / 1F Floor	6" (150mm)	20ft × 30ft (incl. terrace cantilever)	Two-way + cantilever reinforcement	Terrace 5ft cantilevered beyond front wall. Extra reinforcement at cantilever.

Stair Slab (sloped)	5" (125mm)	3.5ft × 9ft at 53°	One-way slab, 10mm bars	Rises from partition (floor level) to front wall (roof level). Roof access.
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4.4 Load Path / भार मार्ग

Roof Slab → Beams (Deep beams at z=9 and z=25) → Columns (C1-C8) → Foundation

Water Tank load → Roof Slab → Beams → Columns

Partition wall (4.5" brick) sits on middle deep beam — not a load-bearing wall itself.

Important / ध्यान दें:

- All columns must be continuous from foundation to roof (no breaks at floor level)
- Roof slab must have 1.5% slope toward front for water drainage
- Stair well opening (3ft × 8ft) requires trimmer beams around the opening
- Cantilever (terrace) must have top reinforcement extended 2× cantilever length into slab

5. Door & Window Schedule / दरवाजे और खिड़कियाँ

5.1 Doors / दरवाजे

ID	Location	Size (W × H)	Type	Material	Remarks
D1	Room entry (partition wall, center)	3ft × 7ft	Single leaf, opening inward	Sal wood frame + flush door	Main room door. Opens from balcony into living area.
D2	Bathroom (from room, right side)	2.5ft × 7ft	Single leaf, opening outward	WPC (Water-Proof Composite) frame + WPC door	En-suite. Opens outward into room. Waterproof material.

5.2 Windows / खिड़कियाँ

ID	Location	Size (W × H)	Sill Height	Type	Remarks
W1	Back wall (room) — Left	4ft × 4ft	3ft from floor	Sliding (2 panel, aluminium)	Cross ventilation. Mosquito mesh. Overlooking farm.
W2	Back wall (room) — Right	4ft × 4ft	3ft from floor	Sliding (2 panel, aluminium)	Same as W1. Symmetrical placement.
W3	Left wall (room) — Center	4ft × 4ft	3ft from floor	Sliding (2 panel, aluminium)	Natural light + ventilation from left side.
W4	Right wall (room) — in Living zone	3ft × 4ft	3ft from floor	Sliding (2 panel, aluminium)	Between kitchen and bathroom. Smaller due to kitchen/bath walls.
W5	Bathroom — Right wall (high)	2ft × 2ft	6ft from floor	Top-hung (outward opening)	Ventilation + light. Frosted glass. High placement for privacy.
V1	Kitchen — Back wall (above counter)	3ft × 2ft	4ft from floor	Top-hung / fixed louver	Smoke & heat ventilation. Above kitchen counter.

5.3 Openings / खुले हिस्से

ID	Location	Size	Treatment
O1	Balcony — Front (between C5 & C8)	~18ft × 9ft (full front open)	MS railing (3.5ft high) with vertical bars at 4" spacing. No wall.

O2	Stair well opening	3ft × 8ft	MS railing (3.5ft high) on 3 sides. Open side toward stairs.
O3	Terrace — Outer edge	20ft × 3.5ft (railing height)	MS railing at outer edge. No wall.

Window Notes / खिड़की नोट:

- All aluminium windows: powder-coated dark grey / brown finish
- Glass: 5mm clear float glass (windows), 4mm frosted (bathroom)
- All windows to have mosquito mesh (SS wire, 20 gauge)
- Lintel above every door/window: 9" × 6" RCC, extending 6" beyond each side

6. Material Specifications / सामग्री विवरण

6.1 Structural Materials

Item	Specification	Quantity Basis
Cement	OPC 43 grade (Ultratech / ACC / Ambuja)	Columns, beams, slab
Sand	River sand (zone II), clean & washed	Concrete + mortar
Aggregate (Gitti)	20mm crushed stone (angular)	RCC work
Steel (TMT bars)	Fe-500D grade (Tata Tiscon / SAIL)	All RCC members
Concrete Grade	M25 (columns, beams, slab)	All structural elements
Bricks	First class burnt clay bricks (crushing strength ≥ 75 kg/cm ²)	All walls

6.2 Wall Construction

Wall Type	Thickness	Mortar	Location
External walls	9" (230mm) brick	1:6 (cement:sand)	All outer walls of room
Partition wall (room/balcony)	9" brick on deep beam	1:6	z=9 line
Internal walls	4.5" (115mm) brick	1:6	Kitchen half-wall, bathroom walls
Kitchen half-wall	4.5" brick, 3.5ft height	1:4	Kitchen partition (semi-open)
Parapet (roof)	4.5" brick, 3ft height	1:6	All edges of roof slab

6.3 Finishes / फिनिशिंग

Element	Material	Brand / Type
Room floor	Vitrified tiles (2ft $\times$ 2ft, glossy)	Kajaria / Somany, light shade
Kitchen floor	Anti-skid ceramic tiles (1ft $\times$ 1ft)	Kajaria / Johnson, matte
Bathroom floor	Anti-skid ceramic tiles (1ft $\times$ 1ft, dark)	Johnson / Nitco, heavy anti-skid

Bathroom walls	Glazed ceramic tiles (full height 7ft)	Kajaria / Somany
Kitchen dado	Glazed tiles (up to 4ft from floor)	Match kitchen floor
Balcony floor	Rough textured tiles (non-slip)	Parking tiles / Kota stone
Terrace floor	Rough Kota stone / textured tiles	Natural Kota, rough finish
Internal walls	12mm plaster + POP punning + emulsion paint	Asian Paints Tractor Emulsion / equivalent
External walls	12mm + 8mm (double coat) plaster + weather coat	Asian Paints Apex / equivalent
Ceiling	POP punning (smooth) + white emulsion	—

6.4 Plumbing & Sanitary / प्लंबिंग

Item	Specification
Water supply pipes	CPVC (Astral / Supreme), 1" main + 3/4" branch
Drainage pipes	PVC (SWR type), 4" for WC + 2" for basin/floor
WC	Indian style (Orissa pan), ceramic, white
Wash basin	Wall-mounted ceramic, 18" × 12"
Shower	Wall-mounted, overhead type (no mixer initially)
Kitchen sink	SS single bowl (24" × 18"), with drain board
Water tank	Sintex / equivalent, 1000L, on roof above stair zone
Waterproofing (bathroom)	Dr. Fixit / Fosroc on floor + walls (up to 1ft) before tiling
Waterproofing (roof)	Bituminous coat + China mosaic / cool roof coating

6.5 Electrical / बिजली

Item	Specification
Wiring	Concealed in PVC conduit (25mm), Copper 1.5 sq mm (lights) + 2.5 sq mm (power) + 4 sq mm (AC/geyser)
MCB Box	8-way DB (Havells / Schneider), near main door
Switches	Modular type (Havells Coral / Anchor Roma)
Ceiling fans	2 nos (room: 1, balcony: 1) — 48" blade
Lights	LED panel lights (room, kitchen), bulkhead (bathroom, balcony)

AC provision	1 no. in room (right wall or back wall). 6 sq mm cable + drain pipe sleeve.
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7. Construction Notes / ठेकेदार के लिए निर्देश

7.1 Sequence of Work / काम का क्रम

- Column extension (GF to Roof):** C3, C4, C5, C8 columns must be cast continuously from 1F floor level to roof slab level (12ft). Use column box formwork. Ensure vertical alignment with GF columns using plumb bob.
- Wall construction:** Build external walls (9") first, then partition wall on middle beam, then internal walls (bathroom, kitchen). Leave door/window openings with proper lintel provision.
- Bathroom walls:** Build bathroom walls (4.5") straddling the partition line — 4ft inside room + 4ft into balcony zone. Waterproof the floor BEFORE tiling.
- Kitchen half-wall:** Build 4.5" wall to 3.5ft height only. Cap with 4" slab/granite top.
- Beam & slab formwork:** Erect formwork for roof beams and slab. Middle deep beam (9"×20") is critical — ensure adequate propping below.
- Steel fixing:** Fix reinforcement as per structural drawing. Ensure minimum cover (1.5" for slab, 2" for beams, 2" for columns).
- Slab casting:** Cast entire roof slab in one pour (20ft × 25ft). Vibrate properly. Cure for minimum 14 days.
- Stair slab:** Cast sloped slab (3.5ft × 9ft, 53°) after roof slab has cured. Anchor properly into partition beam and roof slab.
- Waterproofing (roof):** After curing, apply waterproofing treatment. Ensure 1.5% slope toward front with PCC.
- Plastering:** External double coat (12mm + 8mm), internal single coat (12mm). Cure each coat.
- Electrical & plumbing:** Concealed conduit and pipes before plastering. Test plumbing pressure before closing walls.
- Tiling & finishing:** Floor tiles after all wet work complete. Wall tiles in bathroom/kitchen first, then POP punning, then painting.

7.2 Critical Points / महत्वपूर्ण बातें

#	Point	Reason / कारण
1	Columns C5, C8 must extend full height to roof	Without these, roof slab at front has no support. 20ft span cannot be free.
2	Middle deep beam (9"×20") at z=9 is the most important beam	Carries partition wall + room slab + balcony slab loads. Under-sizing will cause deflection.
3	Stair well opening needs trimmer beams	3ft×8ft hole in slab weakens it. Trimmer beams redistribute load around opening.
4	Bathroom waterproofing before any tiling	Leakage into GF is very expensive to fix later. Use membrane + liquid waterproofing.

5	Roof slope must go toward FRONT (south)	Rainwater drains toward road side. No water pooling on roof.
6	Terrace has NO roof above – railing only	Do not build any wall or slab over the 5ft terrace portion. It is the cantilever from GF slab.
7	Sloped stair slab: ensure adequate anchorage	53° is steep. The slab must be properly anchored at both ends (beam at bottom, roof slab at top).
8	All concealed pipes/wires BEFORE plastering	Chasing walls after plaster wastes time and weakens walls.

7.3 Contractor Checklist / चेकलिस्ट

- ☐ Column reinforcement checked and photographed before pouring
- ☐ Beam dimensions verified (especially middle deep beam 9"×20")
- ☐ Slab reinforcement: spacing, cover, overlap length checked
- ☐ Stair well opening size confirmed (3ft × 8ft) with trimmer beams
- ☐ Plumbing pressure test passed (bathroom, kitchen supply lines)
- ☐ Electrical conduit continuity tested before plastering
- ☐ Bathroom floor waterproofing tested (48-hour ponding test)
- ☐ Roof slope verified (water should flow toward front edge)
- ☐ Roof waterproofing applied and tested
- ☐ All windows/doors: lintels cast before wall construction above
- ☐ MS railing welded, painted with red oxide + enamel (2 coats)
- ☐ Water tank platform and pipe connections completed

7.4 Measurements for Material Ordering

Item	Approximate Quantity	Basis
9" Brick walls (1F)	~850 sqft face area	External walls (20+25+20+25)×12 minus openings
4.5" Brick walls	~200 sqft face area	Bathroom walls + kitchen half-wall + parapet
Roof Slab concrete (M25)	~9.5 cu.m (250 cft)	20ft×25ft×0.5ft slab + beams
Steel (roof slab + beams)	~800-1000 kg	Slab + 4 beams + column extensions
Floor tiles	~500 sqft	Room + balcony + terrace
Wall tiles (bathroom + kitchen dado)	~250 sqft	Bathroom full height + kitchen 4ft dado
Plaster (internal + external)	~2000 sqft	All walls both sides + ceiling

Final Note / अंतिम नोट:

यह दस्तावेज़ architectural design reference के लिए है। Structural design (bar bending schedule, exact reinforcement details) अलग से structural engineer द्वारा provide किया जाएगा। किसी भी doubt में engineer से confirm करें।

This document is for architectural design reference. Structural details (BBS, reinforcement) will be provided separately by the structural engineer. Confirm with engineer in case of any doubt.